



Module Title

Cybersecurity and Networking

Internship Assignment Submission

Linux Fundamentals & Terminal Navigation

Student Name: Jiwosh Shrestha

Internship ID: JVL-CS-2026-8990

Role: Cybersecurity Intern

Duration: 3 months

Assignment Submission Date: Thursday, 17th September 2026

Day:9 (Nine)

Contents

1.THEORY	3
1.1 Answer all 20 questions in your own words	3
2. Practical Work	7
2.1 PRACTICAL SCREENSHOTS (from your own Linux system/VM): whoami · hostname · pwd · ls · ls -l · ls -a · cd / · cd ~.....	7

1.THEORY

1.1 Answer all 20 questions in your own words

1.What is Linux?

Linux is an open-source operating system used in servers, computers, networks, and cybersecurity.

2. What is a Linux distribution?

A Linux distribution is a complete operating system built using the Linux kernel and other software.

3. Name four Linux distributions.

Ubuntu, Debian, Fedora, and Kali Linux.

4. What is Kali Linux?

Kali Linux is a Linux distribution mainly designed for cybersecurity and security testing.

5. What is a terminal?

A terminal is a tool that allows users to control Linux using commands.

6. What does the whoami command do?

It shows username of the current user.

7. What does hostname display?

It displays name of the computer.

8. What does pwd mean?

Pwd means Print Working Directory. It shows your current location in the file system.

9. What is the purpose of ls?

Ls shows the files and folders in the current directory.

10. Explain the difference between ls, ls -l and ls -a.

Ls- shows normal files and folders.

Ls -l shows detailed information about them.

Ls -a also shows hidden files.

11. What does cd do?

Cd means Change Directory. It is used to move between folders.

12. What does cd .. do?

It moves one level up to the parent directory.

13. What does cd ~ do?

It takes you to home directory.

14. What does `cd /` do?

It takes you to root directory of Linux filesystem.

15. What is the Linux root directory?

The root directory `/` is the top-level directory of the Linux filesystem.

16. What is the difference between the root directory and the root user?

The root directory `/` is a location in the filesystem, while the root user is the main administrative user.

17. What is an absolute path?

An absolute path gives the complete location of a file or folder starting from `/`.

Example: `/home/student/Documents/report.txt`

18. What is a relative path?

A relative path shows a location based on your current directory.

19. How are hidden Linux files commonly identified?

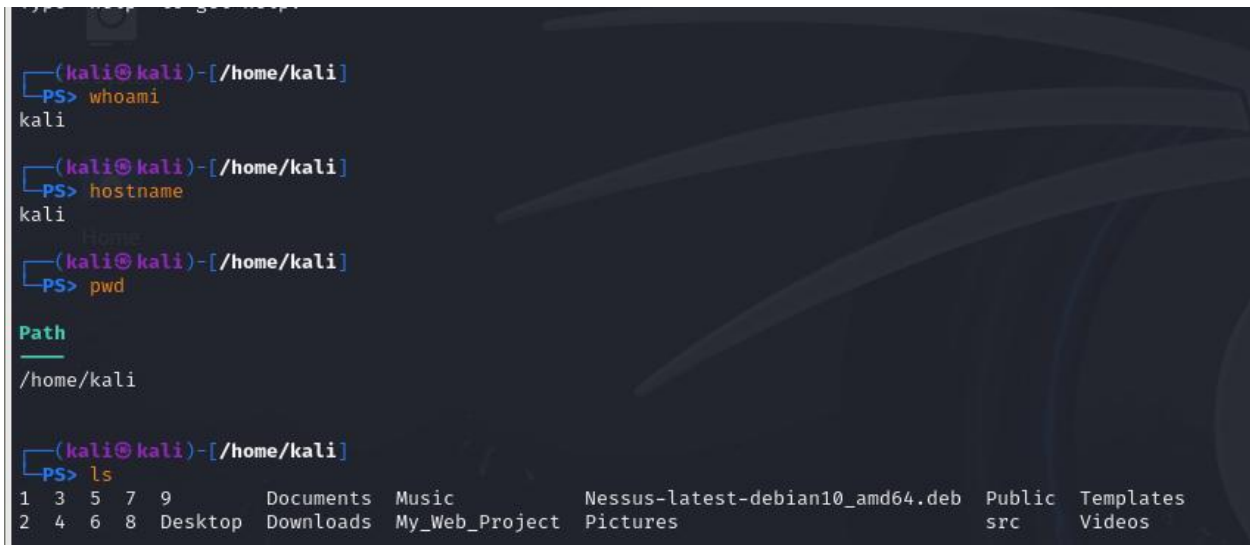
Hidden files usually start with a dot (`.`), such as `.bashrc`. They can be viewed using `ls -a`.

20. Why are Linux skills important for cybersecurity?

Linux skills are important because cybersecurity professionals use the terminals to manage systems, analyze logs, configure systems, investigate problems, and use security tools.

2. Practical Work

2.1 PRACTICAL SCREENSHOTS (from your own Linux system/VM): whoami · hostname · pwd · ls · ls -l · ls -a · cd / · cd ~



A terminal window with a dark background and light-colored text. The prompt is `(kali@kali)-[/home/kali]`. The user enters `PS> whoami` and the output is `kali`. Then the user enters `PS> hostname` and the output is `kali`. Next, the user enters `PS> pwd` and the output is `/home/kali`. Below this, the word `Path` is shown in green with a green underline. Finally, the user enters `PS> ls` and the output is a two-column list of files and directories: `1 3 5 7 9 Documents Music Nessus-latest-debian10_amd64.deb Public Templates` on the first line and `2 4 6 8 Desktop Downloads My_Web_Project Pictures src Videos` on the second line.

```
(kali@kali)-[/home/kali]
PS> whoami
kali

(kali@kali)-[/home/kali]
PS> hostname
kali

(kali@kali)-[/home/kali]
PS> pwd
/home/kali

Path
/home/kali

(kali@kali)-[/home/kali]
PS> ls
1 3 5 7 9 Documents Music Nessus-latest-debian10_amd64.deb Public Templates
2 4 6 8 Desktop Downloads My_Web_Project Pictures src Videos
```

```
(kali㉿kali)-[/home/kali]
```

```
PS> ls -l
```

```
total 67812
drwxrwxr-x 3 kali kali 4096 Jan 21 2026 1
drwxrwxr-x 2 kali kali 4096 Jan 21 2026 2
drwxrwxr-x 2 kali kali 4096 Jan 21 2026 3
drwx----- 2 kali kali 4096 Jan 22 2026 4
drwxrwxr-x 2 kali kali 4096 Jan 21 2026 5
drwxrwxr-x 2 kali kali 4096 Jan 22 2026 6
drwxrwxr-x 2 kali kali 4096 Jan 21 2026 7
drwxrwxr-x 2 kali kali 4096 Jan 21 2026 8
drwxrwxr-x 2 kali kali 4096 Jan 22 2026 9
drwxr-xr-x 2 kali kali 4096 Dec 5 2024 Desktop
drwxr-xr-x 2 kali kali 4096 Dec 5 2024 Documents
drwxr-xr-x 3 kali kali 4096 Jan 21 2026 Downloads
drwxr-xr-x 2 kali kali 4096 Dec 5 2024 Music
drwxrwxr-x 7 kali kali 4096 Jan 22 2026 My_Web_Project
-rw-rw-r-- 1 kali kali 69359018 Mar 12 2026 Nessus-latest-debian10_amd64.deb
drwxr-xr-x 2 kali kali 4096 Jan 22 2026 Pictures
drwxr-xr-x 2 kali kali 4096 Dec 5 2024 Public
drwxrwxr-x 3 kali kali 4096 Jan 22 2026 src
drwxr-xr-x 2 kali kali 4096 Dec 5 2024 Templates
drwxr-xr-x 2 kali kali 4096 Dec 5 2024 Videos
```

```
(kali㉿kali)-[/home/kali]
```

```
PS> ls -a
```

```
. Documents .vboxclient-clipboard-tty7-control.pid
.. Downloads .vboxclient-clipboard-tty7-service.pid
1 .face .vboxclient-display-svgx-x11-tty7-control.pid
2 .face.icon .vboxclient-display-svgx-x11-tty7-service.pid
3 .gnupg .vboxclient-draganddrop-tty7-control.pid
4 .ICEauthority .vboxclient-draganddrop-tty7-service.pid
5 .java .vboxclient-hostversion-tty7-control.pid
6 .local .vboxclient-seamless-tty7-control.pid
7 .mozilla .vboxclient-seamless-tty7-service.pid
8 .msf4 .vboxclient-vmvga-session-tty7-control.pid
9 Music Videos
.bash_logout My_Web_Project .wget-hsts
.bashrc Nessus-latest-debian10_amd64.deb .Xauthority
.bashrc.original Pictures .xsession-errors
.bashrc.save .profile .xsession-errors.old
.cache Public .zsh_history
.config src .zshrc
Desktop .sudo_as_admin_successful
.dmrc Templates
```

```
(kali㉿kali)-[/home/kali]
```

```
PS> cd /
```

```
(kali㉿kali)-[/]
```

```
PS> pwd
```

```
Path
```

```
/
```

```
(kali㉿kali)-[/]
```

```
PS> cd ~
```

```
(kali㉿kali)-[/home/kali]
```

```
PS> pwd
```

```
Path
```

```
/home/kali
```